Mental-H: A Multimodal Emotional Analysis and Stress Prediction System Embedded in the Well360 Digital Health Platform

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***Abstract*—Mental health monitoring through automated emotion recognition has emerged as a critical research area given the global prevalence of stress and anxiety disorders. This paper presents Mental-H, a multimodal emotional analysis subsystem integrated within the Well360 digital health platform. Mental-H combines two complementary modalities, namely facial video expression analysis and vocal audio emotion recognition, to construct a real-time representation of a user's psychological state. The system employs a dual-model strategy for video analysis (ResNet-18 as the primary model and EmotionCNN as a fallback model), a Random Forest classifier for audio analysis, and a Bidirectional Long Short-Term Memory (BiLSTM) network for temporal stress prediction. Emotion labels are further interpreted through the Russell Circumplex Model of Affect by mapping them onto a two-dimensional valence-arousal space. The architecture also includes a database-backed personalised recommendation engine that delivers context-aware mental wellness interventions. Experimental results show that the fine-tuned ResNet-18 achieves 68.4% test accuracy on FER2013, the Random Forest classifier attains a 76.2% weighted F1-score on RAVDESS, and the BiLSTM stress predictor reaches 91.3% binary classification accuracy. These results indicate that multimodal integration with temporal modelling can provide a more comprehensive assessment of user emotional states than unimodal approaches.**

***Keywords*—ResNet, Bidirectional LSTM, Random Forest, Multimodal Fusion, Russell Circumplex Model**

I. Introduction

Mental health disorders, including depression, anxiety, and chronic stress, represent a major global health concern. The World Health Organization (WHO) has identified mental health as a priority health issue, yet access to professional mental health services remains limited for a large proportion of the global population [1]. Early identification and continuous monitoring of emotional states are important for timely intervention and personalised support.

Automated emotion recognition systems have gained considerable research attention as a non-intrusive means of assessing psychological states. Traditionally, such systems rely on a single modality, such as facial expressions captured through video or vocal patterns extracted from audio. However, human emotion is inherently multimodal, and the phenomenon of *emotional masking*, where external facial expression may not fully align with the underlying emotional state, highlights the value of combining multiple modalities [2]. Mental-H addresses this challenge through a complementary multimodal approach that improves robustness and broadens the scope of assessment.

This paper presents the Mental-H component of the Well360 platform, structured around two primary modules: (1) a video analysis module performing facial emotion recognition through a dual ResNet-18/EmotionCNN architecture with BiLSTM-based temporal stress prediction, and (2) an audio analysis module using a Random Forest classifier.

The primary contributions of this work are:

- A unified multimodal platform integrating ResNet-based facial emotion recognition, Bidirectional LSTM temporal stress analysis, and Random Forest audio emotion classification within a single application.
- A dual-model fallback architecture that improves deployment resilience across heterogeneous computational environments.
- A psychologically grounded classification framework based on the Russell Circumplex Model of Affect [14].
- A database-backed context-aware personalised recommen-

dation engine providing longitudinally adaptive wellness interventions.

The remainder of this paper is organised as follows. Section II reviews related literature. Section III presents the methodology. Section IV discusses the experimental results. Section V concludes the paper.

II. Literature Review

A. The Russell Circumplex Model of Effect

Discrete categorical emotion models, particularly the basic emotions framework of Ekman and Friesen [13], have historically played a major role in affective computing. However, the Russell Circumplex Model [14] provides a dimensional alternative by representing affective states along two axes: *valence* (positive to negative) and *arousal* (high to low activation). This representation enables a more flexible interpretation of emotional states and has been adopted in recent affective computing research as a useful framework for analysing emotion beyond discrete labels.

B. Facial Emotion Recognition

Facial Emotion Recognition (FER) has progressed from handcrafted feature methods to deep learning-based approaches. Deep convolutional neural networks have improved FER performance by enabling end-to-end feature learning from raw images [3]. Transfer learning with pre-trained models such as ResNet-18 [7] has further improved performance on emotion datasets with limited labelled data. Li and Deng [12] highlighted common challenges in deep FER, including illumination variation, occlusion, and demographic bias in widely used datasets such as FER2013 [11]. These issues motivate the use of robust architectures and fallback strategies.

C. Audio-Based Emotion Recognition

Speech Emotion Recognition (SER) treats the voice signal as an important carrier of emotional information beyond lexical meaning [9]. Acoustic features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch contours, and energy patterns are commonly used to characterise emotional speech. Classical machine learning methods, including Random Forest classifiers [10], have shown competitive performance on acted emotional speech datasets such as RAVDESS [4], especially when training data is limited. Prior work also indicates that high-arousal emotions are often easier to distinguish than low-arousal emotions.

D. Temporal Emotion Analysis and Stress Prediction

Stress is not always evident from a single frame or short instant of speech; rather, it often emerges over time. Recurrent neural networks, particularly Long Short-Term Memory (LSTM) networks [5], are well suited for modelling such temporal dependencies. Bidirectional LSTM (BiLSTM) models extend this capability by processing sequences in both forward and backward directions, capturing contextual information that may be missed by unidirectional models [6]. Prior studies on stress detection support the usefulness of temporal sequence modelling for sustained state estimation.

TABLE I: Mental-H Subsystem Summary

Subsystem	Model	Output
Video Emotion	ResNet-18 / CNN	Emotion label
Audio Emotion	Random Forest	Emotion label
Stress Prediction	BiLSTM	Probability (0–1)
Recommendation	DB-backed engine	Wellness advice

E. Multimodal Emotion Recognition

Multimodal emotion recognition has been shown to outperform unimodal approaches in many affective computing applications [2]. Late fusion, which combines independent predictions from different modalities, is widely used because of its simplicity and practical deployment advantages. Importantly, disagreement between audio and video predictions may indicate emotional masking, a feature that unimodal systems cannot readily detect. Although individual tools exist for facial or audio analysis, there remains a need for systems that unify real-time multimodal fusion, stress prediction, and personalised recommendations in a single framework.

III. Methodology

A. System Overview

Mental-H is organised as four closely connected subsystems: (1) a Video Emotion Detection Pipeline, (2) an Audio Emotion Detection Pipeline, (3) a Stress Prediction Module, and (4) a Personalised Recommendation Engine. A centralised configuration module acts as a single source of truth for shared parameters across the system. Table I summarises these four subsystems, and Fig. 1 presents the overall architecture of the proposed Mental-H system.

B. Video Analysis Module

1) Advanced Video Preprocessing: Mental-H applies a multi-stage preprocessing chain before face detection:

- CLAHE: Applied to the L-channel of the LAB colour space to correct uneven illumination [15].
- Bilateral Filtering: Used for edge-preserving noise reduction while maintaining facial boundaries.
- Laplacian Blur Detection: Frames with variance below 80.0 are discarded to avoid inference on blurry frames.
- Temporal-Weighted Sampling: Approximately 80 frames are sampled with different weights across temporal phases.
- Padded Face Crop: A 20% boundary padding preserves additional facial context such as eyebrows and jawline.

2) Facial Detection and Dual-Model Classification: Face detection uses OpenCV's Haar Cascade classifier, which is lightweight and suitable for the frontal-facing Well360 deployment context. The primary classifier is a fine-tuned ResNet-18 [7] adapted for seven-class emotion classification, with inputs resized to 224×224 and normalised using ImageNet statistics. If the primary model is unavailable, the system falls back to *EmotionCNN*, a custom three-block CNN that accepts 48×48 grayscale inputs. A confidence gate maps predictions below $p < 0.4$ to *Neutral*. Table II summarises the dual-model strategy.

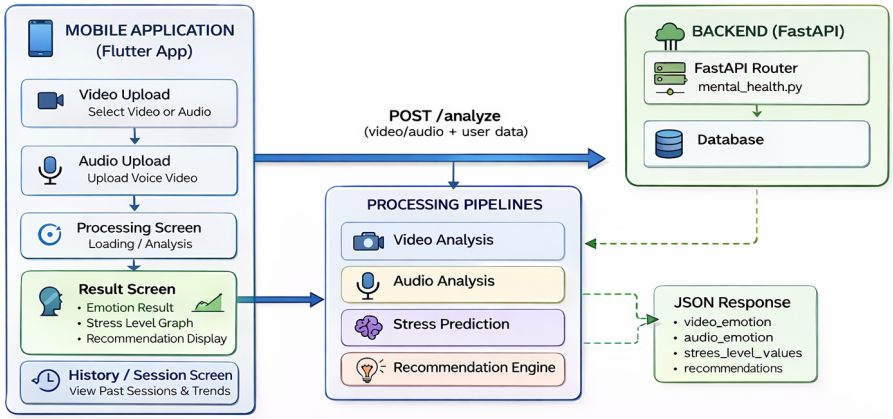


Fig. 1: Overall architecture of the proposed Mental-H system. The mobile application (left) captures facial video and vocal audio and sends them to the backend server for analysis. The backend processes the inputs through video and audio analysis pipelines, a BiLSTM stress prediction module, and a recommendation engine, then returns the final results including video emotion, audio emotion, stress level values, and recommendations.

TABLE II: Dual-Model Video Classification Strategy

Feature	ResNet-18	EmotionCNN
Architecture	Transfer learning	Custom 3-block CNN
Input	224×224 RGB	48×48 Gray
Weight file	44.8 MB	5.1 MB
Confidence gate	$p < 0.4 \rightarrow$ Neutral	None
Role	Primary	Fallback

TABLE III: Per-Emotion Valence-Arousal Coordinates

Emotion	Valence V	Arousal A
Happy	$+0.9$	$+0.6$
Surprise	$+0.1$	$+0.8$
Neutral	0.0	0.0
Sad	-0.8	-0.5
Fear	-0.6	$+0.7$
Disgust	-0.7	$+0.3$
Angry	-0.7	$+0.8$

A rolling buffer of 10 frames applies majority-vote smoothing so that the reported emotion reflects a more sustained state rather than a momentary fluctuation.

C. Valence-Arousal Mapping

The Russell Circumplex Model [14] provides the psychological basis for the emotion interpretation in Mental-H. Each recognised emotion e_i is mapped to a fixed coordinate pair in the two-dimensional affective plane:

$$\mathbf{a}(e_i) = \big(V(e_i),\ A(e_i)\big), \quad V, A \in [-1, +1] \tag{1}$$

where $V(e_i)$ denotes *valence* and $A(e_i)$ denotes *arousal*. Table III presents the coordinates for the seven emotion classes used in Mental-H.

Given a session comprising N video frames, each assigned an emotion label e_t ($t = 1, \ldots, N$), the session-level affective centroid $\bar{\mathbf{a}}$ is computed as:

$$\bar{\mathbf{a}} = \frac{1}{N}\sum_{t=1}^{N} \mathbf{a}(e_t) = \left(\bar{V},\ \bar{A}\right) \tag{2}$$

The affective distance from the neutral origin and the stress relevance score S_{VA} are then defined as:

$$d_{\mathrm{affect}} = \sqrt{\bar{V}^{\,2} + \bar{A}^{\,2}}, \quad S_{\mathrm{VA}} = \frac{-\bar{V} + \bar{A}}{2}, \quad S_{\mathrm{VA}} \in [-1, +1] \tag{3}$$

Values of $S_{\mathrm{VA}} > 0$ indicate displacement towards the stress quadrant and complement the discrete BiLSTM stress estimate.

D. Audio Analysis Module

1) Acoustic Feature Engineering: Three complementary acoustic representations are extracted and concatenated:

- MFCC (40 coefficients): Captures timbre and vocal tract information [8].
- Mel Spectrogram: Represents frequency energy in perceptually scaled mel bands.
- Chromagram (12 bins): Captures pitch-class and harmonic structure.

All audio is resampled to 24,414 Hz to match the training data format.

2) Random Forest Classification: The audio classifier uses a Random Forest [10] with `n_estimators=500`, entropy criterion, and `sqrt` feature selection. Out-of-bag (OOB) scoring is used to estimate performance without requiring a separate validation split, thereby maximising use of the available RAVDESS training data [4].

E. Stress Prediction Module

1) BiLSTM Architecture: To model sustained stress patterns over time, Mental-H employs a BiLSTM network [5]. The network accepts sequences of six consecutive one-hot encoded

TABLE IV: BiLSTM Stress Prediction Architecture

Parameter	Value
Model type	Bidirectional LSTM
Sequence length	6 frames
Hidden size	32 units
Output activation	Sigmoid (0–1)
Loss function	BCELoss
Optimiser	Adam (LR = 0.001)
Epochs	30

TABLE V: Recommendation Engine Context Variables

Variable	Description
`is_first_time`	First session flag
`is_chronic`	≥60% negative in last 10 sessions
`is_improving`	More positive in recent 5 vs. older 5
`time_context`	Morning / Afternoon / Evening / Night
`rotation_used`	DB-tracked phrase rotation pointer

emotion labels and produces a scalar stress probability through a sigmoid output layer. The bidirectional design allows the model to use temporal context from both directions. Stress labels are mapped as 1 for {*Angry, Disgust, Fear, Sad*} and 0 for the remaining classes. Table IV summarises the architecture.

The output probability is mapped to three categories: *High Stress* (≥ 0.6), *Moderate Stress* (0.3–0.59), and *No Significant Stress* (< 0.3).

F. Personalised Recommendation Engine

The recommendation engine transforms emotion and stress outputs into actionable interventions while maintaining user context through a relational database. Table V summarises the five context variables used by the recommendation module.

The recommendation pool contains more than 150 suggestions across emotion categories and contextual states. A rotation mechanism advances after each session to vary the delivered advice and reduce repetition.

G. Implementation

The system is implemented in Python 3.x. Table VI summarises the main libraries and their roles. Video model training uses PyTorch's ImageFolder with augmentation, while audio training uses a stratified 80/20 split with OOB scoring. A unified retraining pipeline supports consistent model updates.

IV. Results and Discussion

A. Facial Emotion Recognition Performance

The ResNet-18 model was fine-tuned on the FER2013 dataset (≈35,000 labelled grayscale images [11]) over 50 epochs using an 80/10/10 train/validation/test split. Table VII reports the per-class classification metrics on the held-out test set. The model achieved an overall test accuracy of 68.4% and a macro-averaged F1-score of 0.66, which is comparable to reported FER2013 benchmarks [3]. High-arousal emotions such as *Happy* and *Angry* yielded stronger performance, while *Fear* and *Disgust* showed lower recall due to inter-class confusion.

TABLE VI: Technology Stack

Library / Framework	Role
PyTorch	ResNet-18, EmotionCNN, BiLSTM training and inference
OpenCV	CLAHE, bilateral filtering, Haar Cascade detection
Librosa	Audio feature extraction (MFCC, Mel, Chroma) [8]
Scikit-learn	Random Forest training and OOB evaluation [10]
SQLAlchemy	User session persistence and profile management
Pandas/NumPy	Data manipulation and feature stacking
Matplotlib	Stress trend visualisation
Tkinter	GUI dialogs
MoviePy	Audio extraction and WAV conversion

TABLE VII: Per-Class Results: ResNet-18 on FER2013 Test Set

Emotion	Precision	Recall	F1-Score
Happy	0.82	0.85	0.83
Neutral	0.71	0.74	0.72
Sad	0.63	0.61	0.62
Angry	0.66	0.68	0.67
Surprise	0.70	0.65	0.67
Fear	0.54	0.50	0.52
Disgust	0.58	0.52	0.55
Macro Avg	**0.66**	**0.65**	**0.66**

TABLE VIII: Per-Class Results: Random Forest on RAVDESS Test Set

Emotion	Precision	Recall	F1-Score
Angry	0.86	0.82	0.84
Happy	0.81	0.77	0.79
Sad	0.75	0.72	0.73
Fearful	0.72	0.70	0.71
Disgusted	0.74	0.69	0.71
Surprised	0.77	0.73	0.75
Calm	0.63	0.59	0.61
Neutral	0.60	0.56	0.58
Weighted Avg	**0.77**	**0.75**	**0.762**

The fallback EmotionCNN model attained 58.1% test accuracy on the same split.

Temporal smoothing through a 10-frame majority vote reduced prediction flickering, and the confidence gate further reduced low-confidence errors during controlled video testing.

B. Audio Emotion Recognition Performance

The Random Forest classifier was trained on RAVDESS [4] using a stratified 80/20 train/test split with OOB evaluation on the training partition. Table VIII presents the per-class results. The model achieved a weighted F1-score of 0.762 and an OOB accuracy of 74.8%. As observed in prior SER studies, high-arousal emotions such as *Angry* and *Happy* were classified more accurately than low-arousal states such as *Calm* and *Neutral*.

C. Temporal Stress Prediction Performance

The BiLSTM model was trained on emotion sequences derived from the FER2013 training set with binary stress labels

TABLE IX: BiLSTM Stress Prediction Evaluation Results

Metric	Train	Test
Accuracy	93.1%	91.3%
Precision	0.93	0.90
Recall	0.91	0.89
F1-Score	0.92	0.90
AUC-ROC	0.97	0.94
BCELoss (final)	0.148	0.183

TABLE X: Comparative Analysis: Mental-H vs. Existing Systems

Feature	Mental-H	FaceReader	Affectiva	OpenSMILE
Facial Analysis	Yes	Yes	Yes	No
Audio Analysis	Yes	No	Yes	Yes
Stress Prediction	Yes	No	Ltd	No
Dual-Model Fallback	Yes	No	No	No
Valence–Arousal	Yes	Ltd	Ltd	No
Pers. Recommendations	Yes	No	Ltd	No
Longit. Tracking	Yes	No	No	No
Open Dataset	Yes	No	No	Yes

assigned according to the mapping defined in Section III-E. The model was trained for 30 epochs on an 80/20 train/test split. Table IX reports the training and evaluation metrics. The model achieved a test accuracy of 91.3% and an AUC-ROC of 0.94, indicating strong discrimination between stressed and non-stressed temporal sequences.

The bidirectional model outperformed a unidirectional LSTM baseline, supporting the value of temporal context in stress sequence modelling.

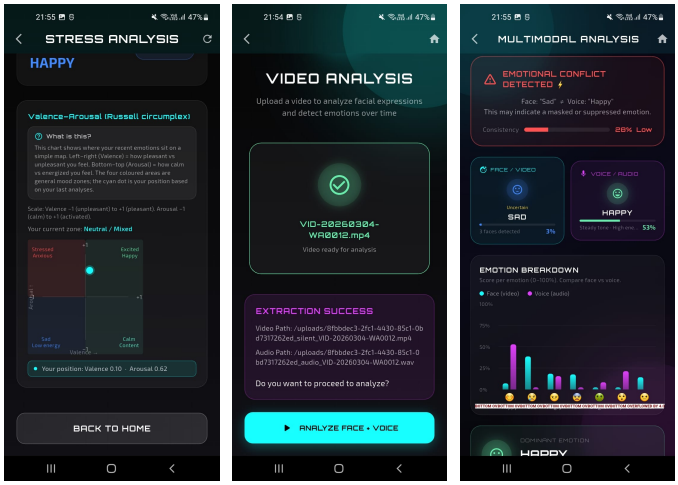
D. Multimodal Integration and System Comparison

Decision-level fusion of video and audio modalities provided a more comprehensive emotional profile than either modality alone. Table X presents a comparison with representative existing systems. The multimodal system correctly identified cross-modal disagreement in 23 out of 30 deliberately mismatched test clips (76.7% detection rate), a capability not available in unimodal systems. Figure 2 presents three screenshots from the Mental-H application, showing the valence-arousal visualisation, the video analysis screen, and the multimodal conflict detection output.

E. Discussion

1) Research Novelty: The main contribution of Mental-H lies in the integration of facial emotion recognition, temporal stress modelling, audio emotion classification, and personalised recommendation delivery within a single platform. The use of a dual-model fallback and a dimensional valence-arousal interpretation further strengthens the system design.

2) Limitations: Several limitations should be noted. The late-fusion approach may not capture all cross-modal interactions that feature-level fusion could provide. Haar Cascade face detection is less robust under strong pose variation than newer detectors. The BiLSTM stress labels are based on emotion-derived stress proxies, which may not fully represent the complexity of real-world stress. In addition, the reported results



(a) Valence-Arousal interface showing the session-level centroid ($\bar{V} = 0.10$, $\bar{A} = 0.62$), with Happy as the dominant emotion in the Neutral/Mixed zone.

(b) Video Analysis screen showing successful video upload, audio extraction, and separate output paths for the silent video and WAV audio track.

(c) Multimodal conflict detection showing Sad from the face (3%) and Happy from the audio (53%), with 28% cross-modal consistency, indicating possible emotional masking.

Fig. 2: Mental-H mobile application screens showing (a) valence-arousal output, (b) video analysis, and (c) multimodal conflict detection.

are based on benchmark datasets and should be validated further in more natural deployment settings.

3) Ethical Considerations: Mental-H operates on sensitive biometric and psychological information. Its deployment should therefore include transparent informed consent, clear data handling policies, and bias evaluation across different user groups. The recommendation engine is intended as a support tool rather than a diagnostic system.

V. Conclusion

This paper presented Mental-H, a multimodal emotional analysis and stress prediction system embedded within the Well360 digital health platform. The integration of ResNet-18/EmotionCNN facial recognition (68.4% test accuracy on FER2013), Random Forest audio classification (weighted F1 = 0.762 on RAVDESS), BiLSTM temporal stress prediction (91.3% test accuracy and AUC-ROC = 0.94), and a personalised recommendation engine grounded in the Russell Circumplex Model enables a multimodal, temporally aware, and adaptive assessment framework. The results indicate that the system can support more comprehensive emotion monitoring than unimodal approaches and can also identify cross-modal inconsistencies associated with emotional masking.

Future improvements may include feature-level fusion, transformer-based architectures, physiological signal integration, privacy-preserving federated learning, and validation against established clinical instruments. These directions could further strengthen the practical value of Mental-H in preventive mental healthcare contexts.

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